Synergid cells	Degenerate
Secondary nucleus	Endosperm
Antipodal cells	Degenerate

Endosperm

The primary endosperm nucleus (PEN) divides immediately after fertilization but before the zygote starts to divide, to form the endosperm. The primary endosperm nucleus is the result of triple fusion (two polar nuclei and one sperm nucleus) and thus has 3n number of chromosomes. It is a nutritive tissue and regulatory structure that nourishes the developing embryo. Depending upon the mode of development three types of endosperm are recognized in angiosperms. They are nuclear endosperm, cellular endosperm and helobial endosperm (Figure 1.21).

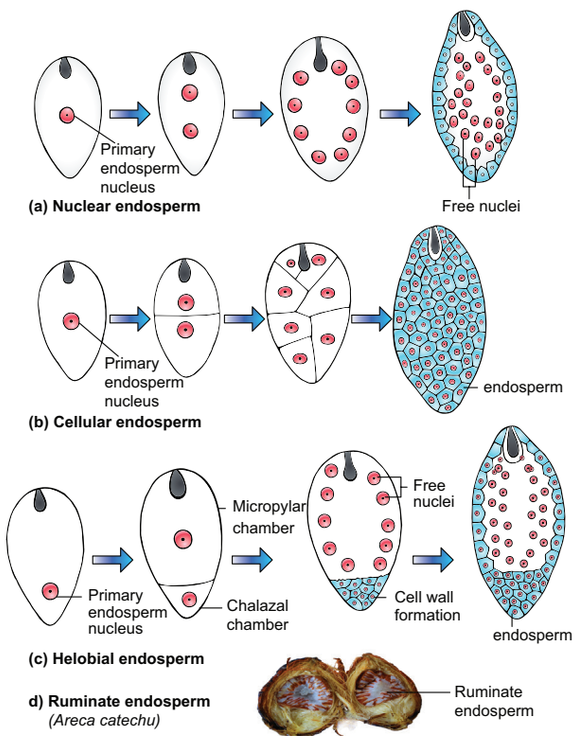


Figure 1.21 Types of Endosperm

1. Nuclear endosperm: In the nuclear type, the Primary endosperm nucleus (PEN) divides into two without any wall formation. The subsequent division of these two nuclei are free nuclear so that the endosperm consists of only free nuclei and cytoplasm around them. The nuclei may either remain free or may become separate by walls in later stages. Examples: *Coccinia, Capsella and Arachis.*



Aleurone tissue consists of highly specialised cells of one or few layers which are found around the endosperm of cereals (barley and maize). Aleurone grain contains sphaerosomes. During seed germination cells secrete certain hydrolytic enzymes like amylases, proteases which digest reserved food material present in the endosperm cells.

2. Cellular endosperm: In this type the primary endosperm nucleus(PEN) divides into 2 nuclei which are immediately followed by a wall formation. Subsequent divisions are also followed by walls. Examples: *Adoxa*, *Helianthus* and *Scoparia*.

3. Helobial endosperm: In helobial type, the primary endosperm nucleus (PEN) moves towards the base of the embryo sac where it divides into two nuclei. These 2 nuclei are separated by a wall to form a large micropylar chamber and a small chalazal chamber. The nucleus of the micropylar chamber undergoes several free nuclear divisions whereas that of the chalazal chamber may or may not divide. Examples : *Hydrilla and Vallisneria*

The endosperms may either be completely consumed by the developing embryo or it may persist in the mature seeds. Those seeds without endosperms are called **non-endospermous** or **ex-albuminous** seeds. Examples: Pea, Groundnut and Beans. Those seeds with endosperms are called **endospermous** or **albuminous** seeds. The endosperms in these seeds supply nutrition to the embryo during seed germination. Examples: Paddy, Coconut and Castor.

Ruminate endosperm: The endosperm with irregularity and unevenness in its surface forms ruminate endosperm (Example: *Areca catechu*). The activity of the seed coat or the endosperm itself results in this type of endosperm. The unequal radial elongation of the layer of seed

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